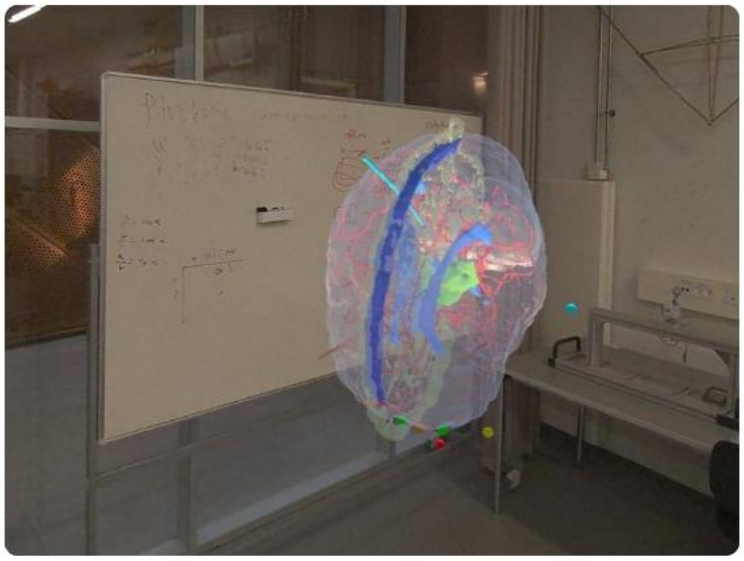
SDU STUDENT RESEARCH PROJECT · TRL 2 TO 3

Curved paths to deep brain targets

Planning smooth, curved paths to deep brain targets while avoiding vessels, ventricles and other no-go structures, then viewing the plan in mixed reality.

See the planner ↗ Watch it in AR ↗

SEEG trajectory planning • Mixed reality • Curved trajectory planning



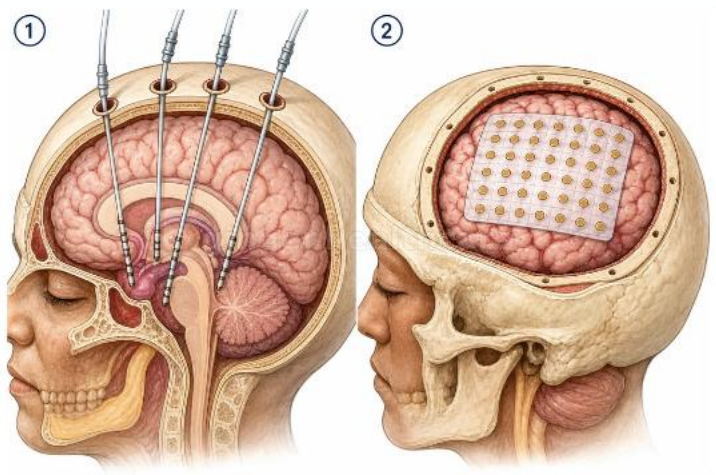
Planned brain model viewed in mixed reality on Meta Quest 3 (passthrough).

EPILEPSY AND SEEG

Finding where seizures begin

About 50 million people live with epilepsy. Roughly one in three do not respond to medication. For them, surgery can stop the seizures, but only if doctors know exactly where they start.

Epilepsy is a chronic brain disease. Seizures are caused by groups of brain cells firing excessive electrical signals in a feedback loop. With the right medication, about 70% of patients become seizure-free. The remaining 30% have drug-resistant epilepsy.
Scalp EEG, MRI, PET and other non-invasive tests can point to a region but often cannot identify it precisely. Stereo-electroencephalography, known as SEEG, closes that gap. Surgeons insert several thin electrodes through small holes in the skull to record brain activity in 3D over days of monitoring.



Invasive EEG can use depth electrodes placed inside the brain or electrode grids positioned on its surface.

50M
people worldwide live with epilepsy

30%
have drug-resistant epilepsy

4 to 18
recording contacts on each electrode

01

Confirm the target

SEEG tests where seizures start and how they spread.

02

Reach deep structures

It records from the hippocampus, insula, cingulate and deep sulci.

03

Protect function

It maps the relationship to movement, speech, vision and memory.

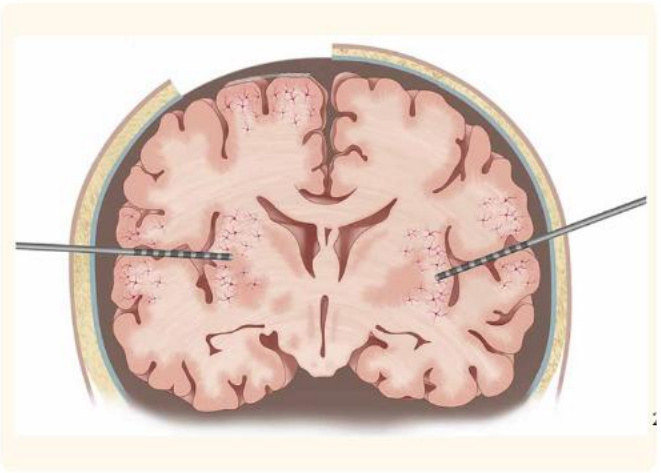
SEEG is a diagnostic decision tool, not a treatment. Its value depends on placing every electrode exactly where it is needed without touching

SEEG is a diagnostic decision tool, not a treatment. Its value depends on placing every electrode exactly where it is needed without touching vessels or critical tissue.

THE PROBLEM

Straight lines are not always the safest way in

Today's robotic SEEG systems place every electrode along a straight line. We explore whether curved trajectories could provide safer routes around critical anatomy.



Standard SEEG electrodes follow fixed straight trajectories from the skull opening to deep recording targets.

WHY CURVATURE MATTERS

A precise straight line can still be blocked

Each standard SEEG electrode enters through a small skull opening and continues along one fixed axis. Robotic guidance makes that line highly accurate, but it cannot curve around a vessel, ventricle or another protected structure.
When no safe straight corridor exists, the plan may require a different entry point or an additional electrode. Curved trajectory planning could follow the anatomy instead.

01

Straight lines only

Systems such as ROSA, Neuromate and iSYS1 align a guide with 1 to 2 mm accuracy, but the electrode itself cannot bend.

02

Deep targets are hard

The hippocampus and other deep structures are surrounded by vessels, ventricles and critical tissue.

03

More holes, more risk

Reaching complex regions can require extra skull entry points, raising the risk of bleeding and infection.

THE PLANNER

Plan a curved path in minutes, not by hand

NeuroSNAKE Planner is a Windows desktop application. It loads a patient-specific 3D brain model, lets the user choose targets, and finds curved paths that satisfy configured anatomical safety limits.

Patient-specific model | No-go zones | A* search | Curved path planning | Configurable safety margin

01

Load the patient

Open a patient folder with colour-coded 3D anatomy built from MRI.

02

Mark safety zones

Block vessels, ventricles and the midline. Add margins and softer penalty regions.

03

Choose targets

Place targets in three slice views. Unsafe points turn red and can snap to safety within 5 mm.

04

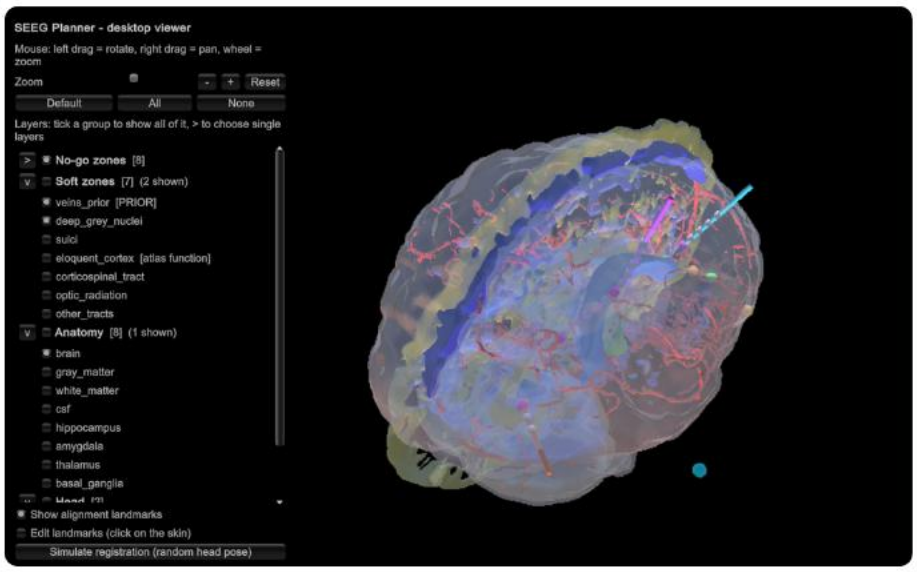
Set constraints

Configure safety margins, entry conditions and planning limits in one editable file.

05

Plan and inspect

Find a coarse route on a 2 mm grid, fit a smooth spline and keep only paths that pass every hard limit.



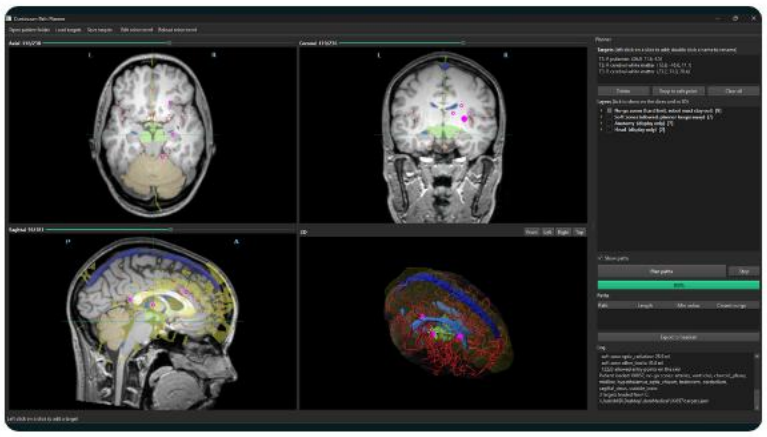
Planned SEEG trajectories shown inside the layered patient model.

READING THE PATH

The stripes mark electrode contacts

The small transverse stripes visible along each coloured trajectory represent the recording contacts on the SEEG electrode. They are positioned at 5 mm intervals along the path.
These contacts record electrical activity from several depths along the planned trajectory during the SEEG examination.

5 mm spacing



Three targets selected across the anatomical views before trajectory planning.



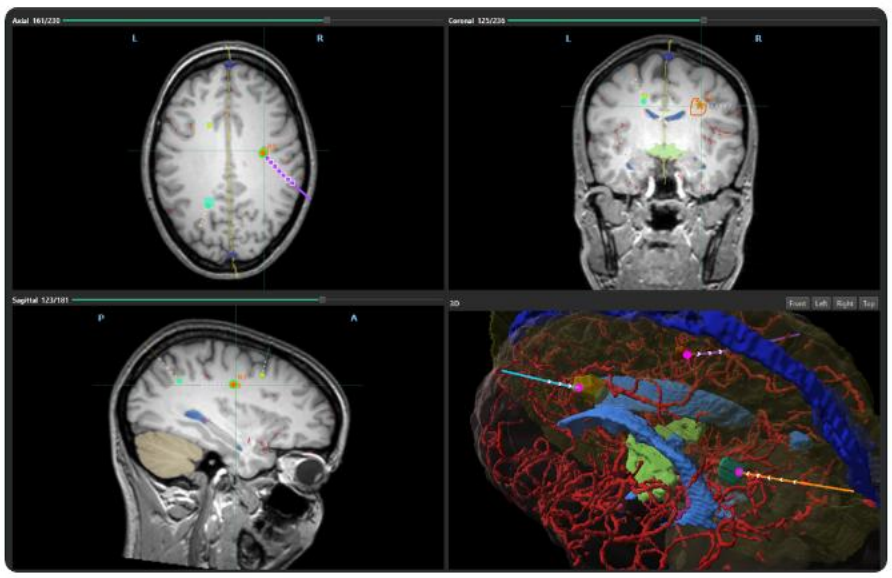
Planned multi-target trajectories shown with anatomical layers and safety structures.

TARGET REGIONS

Define an area, not just a single point

Instead of picking a single point, the doctor marks the area the trajectory should reach. They draw its outline with the mouse on a slice in one view, such as axial, and then in a second view, such as coronal. The program combines the outlines into a 3D region: the part where the outlines overlap, clipped to the brain. The region appears immediately on all slices and in the 3D view. During planning, the path may end anywhere inside the region. The program picks a safe point that is easiest to reach from the skin and lies away from the region's edge, while avoiding all no-go zones. For each path it shows how many electrode contacts lie inside the region. Regions are saved with the plan and shown in the Meta Quest 3 headset.

Draw in two views **Combine into 3D** **Plan to the safest point**



Target regions and planned trajectories shown across axial, coronal, sagittal and 3D views.

PATIENT DATA

From raw scans to a 30-layer patient model

The model is built from a real person's own MRI scans, not from an average brain. One pipeline turns the scans into colour-coded 3D layers, safety maps and a headset scene.

The current test subject is IXI057 from the public IXI dataset. T1-weighted MRI provides brain and head anatomy, TOF-MRA highlights arterial blood, and diffusion MRI shows the direction of white-matter fibres.
Patient data, atlas priors and estimates are labelled separately. Every critical structure receives a distance map in millimetres so the planner can enforce safety margins honestly.

95
brain regions from FastSurfer segmentation

30
anatomical layers in the headset scene

0.78
Dice score for automatic artery extraction

01
Segment anatomy
FastSurfer separates cortical regions, deep structures and ventricles from T1 MRI.

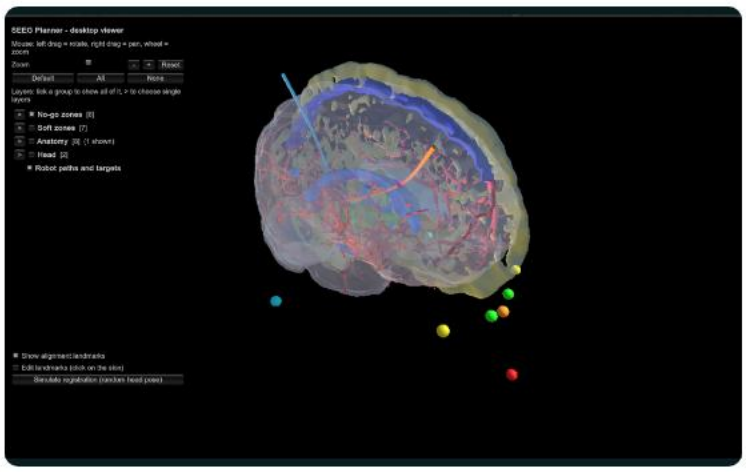
02
Extract vessels
A registered angiography scan and vesselness filtering create the arterial layer.

03
Track fibres
Diffusion MRI produces nine major white-matter tract groups with DIPY.

04
Add safety context
Sulci, midline, eloquent cortex and carefully labelled priors complete the model.

05
Build the scene
Smoothed meshes and distance maps serve both the planner and the headset export.

> Atlas-derived veins remain a soft zone, not a blocked patient-specific structure. The app marks every prior and estimate so it is never presented as measured patient data.



The Unity desktop viewer shows anatomical layers, planned paths and registration landmarks.

MIXED REALITY

See the plan where the patient is

A plan on a flat screen is hard to judge in depth. Our Unity app places the brain, anatomy and planned path as a life-size hologram in the real room.
Planning stays in the desktop app. With one click, it exports the brain layers, groups and trajectories as glTF. Unity loads the scene on Meta Quest 3 in passthrough mode.

- Walk around the plan and inspect depth
- Show or hide skin and anatomical layer groups
- Register the hologram using eight automatically selected facial landmarks
- Grab and reposition the model by hand

Python planner → glTF → Unity 6 · OpenXR · Meta Quest 3



THE HEADSET

Meta Quest 3 with colour passthrough

The first demo runs on Meta Quest 3 connected to a PC. Its external cameras show the real room while Unity places the planned anatomy and paths into that view. Touch Plus controllers switch layers, move the model and collect registration landmarks.
The hologram begins about half a metre in front of the user. A short help guide appears above the left controller. The same Unity scene also runs on a normal desktop for users without a headset.

Registration is approximate because the demo uses a public test brain. The long-term target is an optical see-through headset such as HoloLens 2 or Magic Leap 2.

STATUS AND NEXT STEPS

Where we are, and what comes next

NeuroSNAKE is a student research prototype at technology readiness level 2 to 3. It shows that curved SEEG paths can be planned, checked and visualised. It is not a medical device.

CURRENT LIMITATIONS

- Simplified path models without tissue deflection
- Testing on public brain scans rather than real patients
- Planning time above the 30-second target
- No side-by-side comparison with straight SEEG yet

WHAT COMES NEXT

- Real patient MRI and CT angiography
- Comparison with straight SEEG on the same anatomy
- A standalone headset build without a PC
- More precise hologram registration
- Feedback from neurosurgeons

AUTHOR

Created at the University of Southern Denmark

NeuroSNAKE was developed as a student research project in planning software, medical imaging and mixed reality.



Jakub Adam Banaszek

Planning software, experiments and project development

Supervised by **Di Wo**

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